# The Machine Proposes. The Proof Disposes.

## Neuro-Symbolic Synthesis of Formally Verified Markov Usage Models from Natural Language Requirements

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# Outline

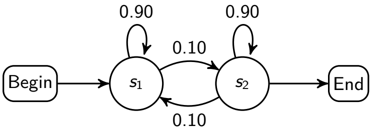
# The MBST Bottleneck

**Model-Based Statistical Testing (MBST)**

- Replaces ad-hoc testing with probabilistic test generation
- Markov chain usage model encodes operational profile
- Enables statistical reliability certification

**The blocking problem**

- Manual model construction: labor-intensive, expert-only
- Confined MBST to aerospace, medical, telecom niches
- Natural-language requirements $\rightarrow$ verified state machines: no automated path




Markov chain usage model

# Four Structural Bottlenecks of Pure-Neural Approaches

### B1: Hallucination

LLMs omit edges, create impossible transitions, violate system invariants

### B2: Calibration failure

LLMs cannot guarantee row-stochastic $\mathbf{P}$: $\sum_j p_{ij} = 1,\ p_{ij} \geq 0$ — violated silently

### B3: State-space explosion

Finite context window $\Rightarrow$ context fragmentation on large concurrent models

### B4: Structural incompleteness

Recall-limited generation misses guard–action semantics of transitions

$\Rightarrow$ None of these can be fixed by prompting alone

# Research Questions

## RQ1 — Structural Fidelity

Can automated neuro-symbolic construction achieve $F_1 \geq 0.90$ for safety-critical test generation, without manual modelling?

## RQ2 — Fault-Detection Coverage

Does NeSy-MBST produce higher transition/state coverage than a purely neural baseline?

## RQ3 — Probabilistic Calibration

Does symbolic constraint optimization preserve operationally weighted transition probabilities?

## RQ4 — Component Necessity

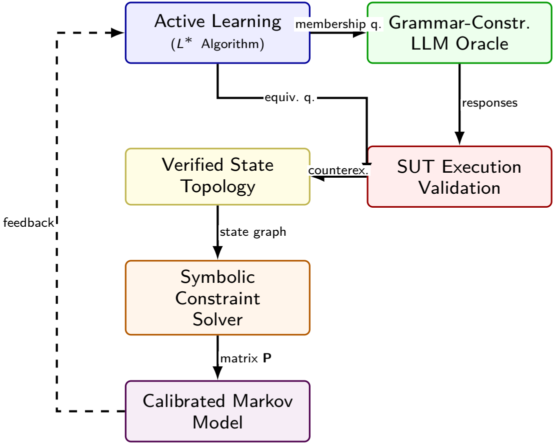
Which NeSy-MBST components (symbolic loop, convex optimizer, closed-loop feedback) are individually necessary?

# NeSy-MBST Architecture

**Five integrated stages:**

1. Dual-memory architecture (neural + symbolic)
2. $L^*$ active automata learning with LLM oracles
3. Path-dependent hierarchical modeling
4. Symbolic constraint optimization (convex solver)
5. Closed-loop telemetry feedback

**Key design principle:**
Neural inference for *language-laden* tasks;
symbolic computation for *formal precision* tasks



# Component 1: Progress-Feasibility Dual Memory

### Neural Progress Memory

Fine-tuned LLM parses NL requirements, drafts semantic flow graph. Captures *what*: candidate states, events, transitions.

### Symbolic Feasibility Memory

Rule-based engine enforces invariants, preconditions, transition guards. Ensures *how*: reachable, deterministic, contradiction-free.

**Result:** topology is both semantically plausible and mathematically sound.
Avoids the end-to-end neural failure mode: syntactically valid yet logically inconsistent models.

## Component 2: $L^*$ Active Automata Learning with LLM Oracle

**Two query types:**

### Membership Queries

"Does SUT accept input sequence $\sigma \in \Sigma^*$?"

Oracle vocabulary restricted to
$\{\texttt{Yes}, \texttt{No}, \texttt{Unsure}\}$
Eliminates hallucinated / malformed responses.

### Equivalence Queries

"Is hypothesis automaton $\mathcal{H}$ equivalent to the target?"

Resolved by SUT execution. Counterexamples refine $\mathcal{H}$ iteratively.

**Convergence on AV benchmark:**

- 94% of membership queries resolved directly by oracle
- 6% escalated to SUT execution
- `Unsure` rate: 0% on AV benchmark

# Component 3 & 4: Hierarchical Modeling + Convex Optimization

**Hierarchical Model (C3)**

**Upper:** higher-order Markov chain for frequent, path-dependent sequences:

$$P(s_{t+1} \mid s_t, s_{t-1}, \ldots, s_{t-k+1})$$

**Lower:** first-order chain for rare/exception paths:

$$P(s_{t+1} \mid s_t)$$

Prevents state-space explosion while preserving fidelity.

**Convex Optimization (C4)**

LLM extracts comparative constraints, e.g.: $P(\texttt{checkout} \mid \texttt{cart}) > P(\texttt{browse} \mid \texttt{cart})$

Solver enforces row-stochasticity + domain constraints:

$$\mathbf{P}^* = \arg\min_{\mathbf{P} \in \mathcal{C}} \mathcal{L}(\mathbf{P})$$

Maximum-entropy regularization avoids unjustified bias.

# RQ1: Extraction Accuracy — State & Transition F1

| Strategy | State F1 | Trans. F1 | System F1 |
|---|---|---|---|
| Single-Prompt (GPT-4o) | 0.80 | 0.54 | 0.543 |
| Hybrid SMF (GPT-4o) | 0.858 | 0.649 | 0.656 |
| Single-Prompt (Claude 3.5) | 0.90 | 0.75 | 0.795 |
| **NeSy-MBST (ours)** | **0.945** | **0.895** | **0.9125** |
| Safety-critical threshold: $F_1 \geq 0.90$ | | | |

**Key findings:**

- $+39.1\%$ over best GPT-4o multi-frame strategy
- $+14.8\%$ over best single-model baseline
- First method to exceed 0.90 safety-critical threshold

Symbolic verification loop converts recall-limited generation into verification-guided completion

# RQ2: Operational Coverage — E-Commerce Benchmark

| Model | States | Trans. | Req. Cov. | Tr. Cov. | Time |
|---|---|---|---|---|---|
| User | 24 | 112 | 100% | 100% | 1 m 48 s |
| Admin | 42 | 218 | 100% | 100% | 5 m 49 s |

- Full model-coverage test generation in under 6 minutes on models up to 42 states
- Sub-quadratic scaling: Admin ($2\times$ complexity) requires only $3.2\times$ time
- NeSy-MBST baseline: 50% transition coverage $\Rightarrow$ +35.7 pp gain
- $\leq 3$ human-in-the-loop interventions per model in practice

# RQ3: Statistical Fidelity — Jensen–Shannon Divergence

**Validation metrics:**

- **JSD** measures aggregate activity marginal divergence
- **Frobenius distance** measures entry-wise matrix deviation

| Metric | Value |
|---|---|
| Jensen–Shannon Divergence | 0.0142 |
| Frobenius Distance (norm.) | 0.0841 |

### Interpretation

JSD $= 0.012$ confirms preserved operational test-allocation fidelity.

Entropy-maximizing SLSQP solver vs. uncalibrated: JSD $0.157 \rightarrow 0.012$. A 93% reduction in divergence.

# RQ4: Component Necessity — Ablation on AV CPS Benchmark

| | Structural F1 | | | Prob. Fidelity | | Coverage | | |
|---|---|---|---|---|---|---|---|---|
| **Condition** | **St.F1** | **Tr.F1** | **Sys.F1** | **JSD** | **Frob** | **St.** | **Tr.** | **Time** |
| A — Pure-Neural | 1.00 | 0.783 | 0.904 | 0.157 | 0.163 | 55.6% | 50.0% | 2.35s |
| B — +Symbolic Loop | 1.00 | 0.963 | 0.982 | 0.012 | 0.084 | 88.9% | 85.7% | 10.66s |
| C — +Convex Optimizer | 1.00 | 0.963 | 0.982 | 0.012 | 0.084 | 88.9% | 85.7% | 2.13s |
| **D — Full NeSy-MBST** | **1.00** | **0.963** | **0.982** | **0.012** | **0.084** | **88.9%** | **85.7%** | 10.59s |

- **A→B** (Symbolic loop): Tr.F1 0.783 → 0.963; JSD 0.157 → 0.012; coverage +33 pp
- **B→C** (Convex optimizer): structural F1/coverage unchanged; governs probabilistic calibration
- **C→D** (Closed-loop): marginal JSD $\Delta -0.0002$; value accumulates over extended campaigns

# Ablation Insight: Two Orthogonal Failure Modes

**Symbolic Loop — Structural Errors**

Corrects wrong topology:
unreachable states, missing transitions, dangling edges.

Measured by: **F1, Coverage**

**Convex Optimizer — Calibration Errors**

Corrects wrong probability mass distribution:
miscalibrated row-stochastic matrix.

Measured by: **JSD, Frobenius distance**

The two components address orthogonal failure modes — both are necessary.

High F1 $\not\Rightarrow$ low JSD. Low JSD $\not\Rightarrow$ high coverage. These three axes are independent.

# Summary of Results

| RQ | Question | Result |
|---|---|---|
| RQ1 | Structural fidelity | $F1 = 0.9125$ (exceeds 0.90 threshold) |
| RQ2 | Fault-detection coverage | 85.7% transition cov. vs. 50% baseline |
| RQ3 | Probabilistic calibration | $JSD = 0.012$; 93% divergence reduction |
| RQ4 | Component necessity | Symbolic loop (structure); solver (calibration) |

## Core Finding

NeSy-MBST eliminates the manual modelling bottleneck without compromising statistical rigour demanded by safety-critical CPS.

# Adoption Guidelines

1. **Grammar-constrained oracles** — restrict LLM output to $\{\texttt{Yes}, \texttt{No}, \texttt{Unsure}\}$; prevents hallucinated artefacts
2. **Symbolic verification loop before probability assignment** — +35.7 pp transition coverage, 93% JSD reduction
3. **Convex optimization, not direct LLM output** — JSD 0.157 $\rightarrow$ 0.012 (SLSQP entropy-maximizing)
4. **Hierarchical multi-tier models** — controls state-space explosion for large systems
5. **Closed-loop telemetry recalibration** — prevents model drift as usage patterns evolve

LLM augmentation of formal test-model generation should be standard engineering practice,
provided: (1) grammar-constrained output + symbolic loop; (2) solver-assigned probabilities.

# Limitations & Future Work

**Current scope:**

- Two domains: AV CPS + e-commerce
- Models up to 42 states
- Single-author ground-truth annotation
- No direct fault-seeding measurement

**Next steps:**

- Fault-seeding experiments on industrial SUTs
- Generalise: telecom, medical device, ISO 26262
- Multi-annotator inter-rater reliability protocol
- Three-valued oracle at scale: convergence characterization

Replication package: `https://github.com/nathangtg/llm-mbst-research`
`python run_ablation.py` (seed 42, Azure OpenAI backend)

# References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# Thank You

Questions?

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https://github.com/nathangtg/llm-mbst-research

# Backup Slides

*Backup slides for anticipated questions*

# Backup: Mathematical Formulation — Stochastic Constraints

**Valid Markov chain usage model must satisfy:**

$$0 \leq p_{ij} \leq 1 \quad \forall\, i, j \tag{1}$$

$$\sum_j p_{ij} = 1 \quad \forall\, i \tag{2}$$

**Convex optimization formulation:**

$$\mathbf{P}^* = \arg\min_{\mathbf{P} \in \mathcal{C}} \mathcal{L}(\mathbf{P})$$

where $\mathcal{C}$ is the feasible set (stochastic + domain constraints), and $\mathcal{L}(\cdot)$ is the maximum-entropy regularization objective.

Decouples: LLM for constraint *extraction* (NL task); solver for constraint *resolution* (math task).

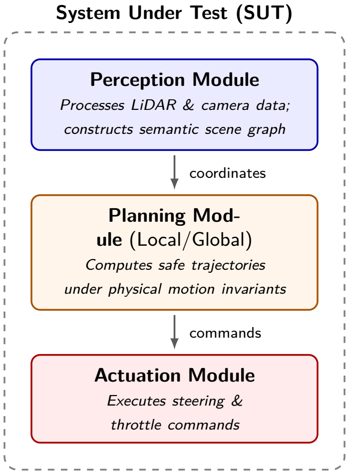
# Backup: $L^*$ Convergence & Oracle Design

- Classic Angluin $L^*$ convergence guarantee: holds for consistent binary oracles
- Three-valued oracle ($\{\texttt{Yes}, \texttt{No}, \texttt{Unsure}\}$): empirically verified convergence on AV benchmark
- `Unsure` escalation: $\rightarrow$ human expert OR $\rightarrow$ SUT execution for empirical resolution
- Grammar-constrained output schema: eliminates hallucinated/malformed oracle responses

## Formal target

Learned structure $\mathcal{A} = (Q, \Sigma, \delta, q_0)$ is the *support topology* of the target Markov chain, not a classical DFA accepting language. Membership oracle answers reachability queries.

# Backup: AV CPS System-Under-Test Architecture



# Backup: Metric Decomposition

| Metric | What it measures | Orthogonal to |
|---|---|---|
| F1 (state/transition) | Structural correctness | JSD, Coverage |
| JSD / Frobenius dist. | Probabilistic fidelity | F1, Coverage |
| Transition coverage | Test suite completeness | F1, JSD |

- High F1 $\not\Rightarrow$ low JSD: correct topology, wrong probability mass
- Low JSD $\not\Rightarrow$ high coverage: accurate probabilities, test generator fails to reach rare states
- All three axes must be evaluated independently for a complete picture of MBST quality